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Géomorphologie et géoarchéologie des littoraux de mer Noire

*Geomorphology and geoarchaeology
of Black Sea coasts*



A. Baralis, A. Bivolaru, N. Marriner, C. Morhange, A. V. Porotov, V. N. Zin'ko dir.



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- 1 - Swallow's Nest, neo-gothic fortress built in 1911-1912 (Crimea)
Source: A. Porotov
- 2 - View of the present Kavarna bay (Bulgaria)
Source: P. Peev
- 3 - Provadia-Solnitsata. Necropolis. Mass grave of murdered men, women and children, buried in a round pit (upper level). Late Chalcolithic, 4500-4200 BC
Tous droits réservés
- 4 - Sambek valley
Source: M. Schlöfel
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Rédacteur en chef: Christophe MORHANGE, morhange@cerege.fr

président de l'association des « Amis de la revue Méditerranée »

Secrétaires de rédaction: Régis DARQUES (géographie humaine), regis.darques@univ-montp3.fr

Edward ANTHONY (géographie physique), anthony@cerege.fr

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SECRÉTARIAT DE RÉDACTION

Presses Universitaires de Provence

Maison de la Recherche - 29, avenue Robert-Schuman - F 13621 Aix-en-Provence cedex 1

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Geoarchaeology of Phasis (Georgia)

Géoarchéologie du Phasis (Géorgie)

Vakhtang LICHELI

Institute of Archaeology, Faculty of Humanities
Tbilisi, Georgia.
vakhlich@hotmail.com

Abstract – The City of Phasis is located on the East coast of the Black Sea (Georgia). The location of the biggest and most successful of the colonies established by Ionian Greeks is still not known, a subject that has attracted interest from scholars for almost two hundred years. Based on written sources, geomorphological and archeological data, it is assumed that, due to tectonic and geomorphological changes, part of the ancient settlement/cultural layers dating back to the 1st mill. BC are below current sea level. These two changes mean that the present landscape is completely different to the 1st mill. BC situation. This observation is significant in resolving Phasis's location. We report that: 1. The Greeks did not establish the city of Phasis adjacent to a lake. 2. The early city was located north of the river. 3. Early Phasis (6th-4th c. BC) is consistent with the settlement of Sagvichio or east of it.

Keywords: Greek colonization, Phasis, earthquake, geomorphology, geology, archaeology

Résumé – La cité de Phasis était disposée sur le littoral oriental de la mer Noire (Géorgie). Cette colonie, la plus importante et la plus prospère parmi celles fondées par les Grecs ioniens, n'a toujours pas été mise au jour en dépit de deux siècles d'efforts infructueux. En se basant sur les sources anciennes, ainsi que sur les données géomorphologiques et archéologiques, on peut présumer que l'ancien établissement se trouve désormais en partie sous l'actuel niveau marin suite aux nombreux changements tectoniques et géomorphologiques qui sont survenus. Ces deux derniers phénomènes ont influencé l'actuel réseau hydrographique qui a été totalement modifié et ne correspond plus à celui du premier millénaire avant J.-C. Cette observation permet de résoudre de façon significative le problème de la localisation de Phasis. On peut dès lors supposer que : 1. les Grecs n'ont pas établi Phasis à proximité d'un lac et que ce dernier n'est pas déterminant pour la localisation de la cité initiale ; 2. celle-ci était située au nord du fleuve ; 3. la première Phasis (6^e-4^e s. av. J.-C.) doit être identifiée avec l'établissement de Sagvichio ou de ses alentours.

Mots-clés : colonisation grecque, Phasis, séisme, géomorphologie, géologie, archéologie

1 - Culture of the Pre-colonial Phasis area

Colchis kingdom, where Miletus established the city of Phasis, was considered as being at the edge of the world both during the Greek colonization and in the following period as well [Socrates: "and then I believe that the earth is extremely large, and that we who live between the pillars of Heracles and Phasis inhabit some small part of it around the sea" (DILLON and GARLAND, 2010)]. At the end of the 2nd mill. BC and at the beginning of the 1st mill. BC Colchis appeared to be one of the most developed state/tribal units in the Caucasus region (LORDKIPANIDZE, 2001). As a result of archeological investigations a most impressive picture of the society's development has been revealed: developed bronze and iron metallurgy, agriculture (producing wheat, millet, fiber and oil plants; wine), traditional architecture (wooden construction), artwork, a developed society and harsh differentiation of property. This is the most significant period in the history of Colchis and is described in ancient Assyrian and Urartian sources as well as in ancient Greek literature and relates to the well-known myth in the Hellenic world regarding the voyage of the Argonauts (LORDKIPANIDZE, 2002). From an archeological point of view several specific events can be selected from this period. One of the most significant phenomenon at the beginning of the 1st mill. BC were collective burial grounds (MIKELADZE, 1995). At Ureki, Nigvziani, Dgvaba, Ergeta (Fig. 1) such burial grounds usually contained large amounts of artifacts and remains of individuals (the quantity of identified individuals varies from dozens to one thousand). The peculiarities of Colchis collective burial grounds are as follows: ritual area,

large-sized collective burial pits, secondary burial canon (PAPUASHVILI, 1989, 1998 et 2004). The finest model of such a burial ground can be described as a table for processing the deceased and a sacrificial area (PAPUASHVILI, 1989). One of the typical examples is the burial ground N4 dated from the second half of the 8th c. BC to the first half of the 7th c. BC which was discovered in Ergeta (PAPUASHVILI, 1989 and 1998). The length of the burial, including the entrance, is 11 m and the width is more than 5 m. Osteological material has not been preserved. There are 596 artifacts in total. Materials include: clay, stone, wood, copper, bronze, iron, lead, silver, gold, semiprecious stones and amber. All of them had different purposes: various types of instruments, tools, agricultural tools, elements related to clothes, necklaces, small-sized sculptures etc. The sculptures of a leopard and a double-headed ram were discovered in this burial ground. In other collective burial grounds (Ureki, Dgvaba, Ergeta) bronze statues of a horsewoman and bulls were discovered (MIKELADZE, 1995). It is significant that relation to the Greek world is shown for the first time in this material, namely two types of statues: Goddesses/horsewomen (which are well known) and bull statues, parallels of which are widespread in the Greek world (LICHELI, 2008).

These statues have special meaning for understanding the pre-colonial cultural conditions in Colchis as well as for investigating the early links between these two regions. Statues of goddesses/ horsewomen are introduced in three regions: the Aegean world, Colchis and North Caucasus (PAPUASHVILI, 2010; for iconography see LORDKIPANIDZE, 2002 and MIKELADZE, 1995). There are

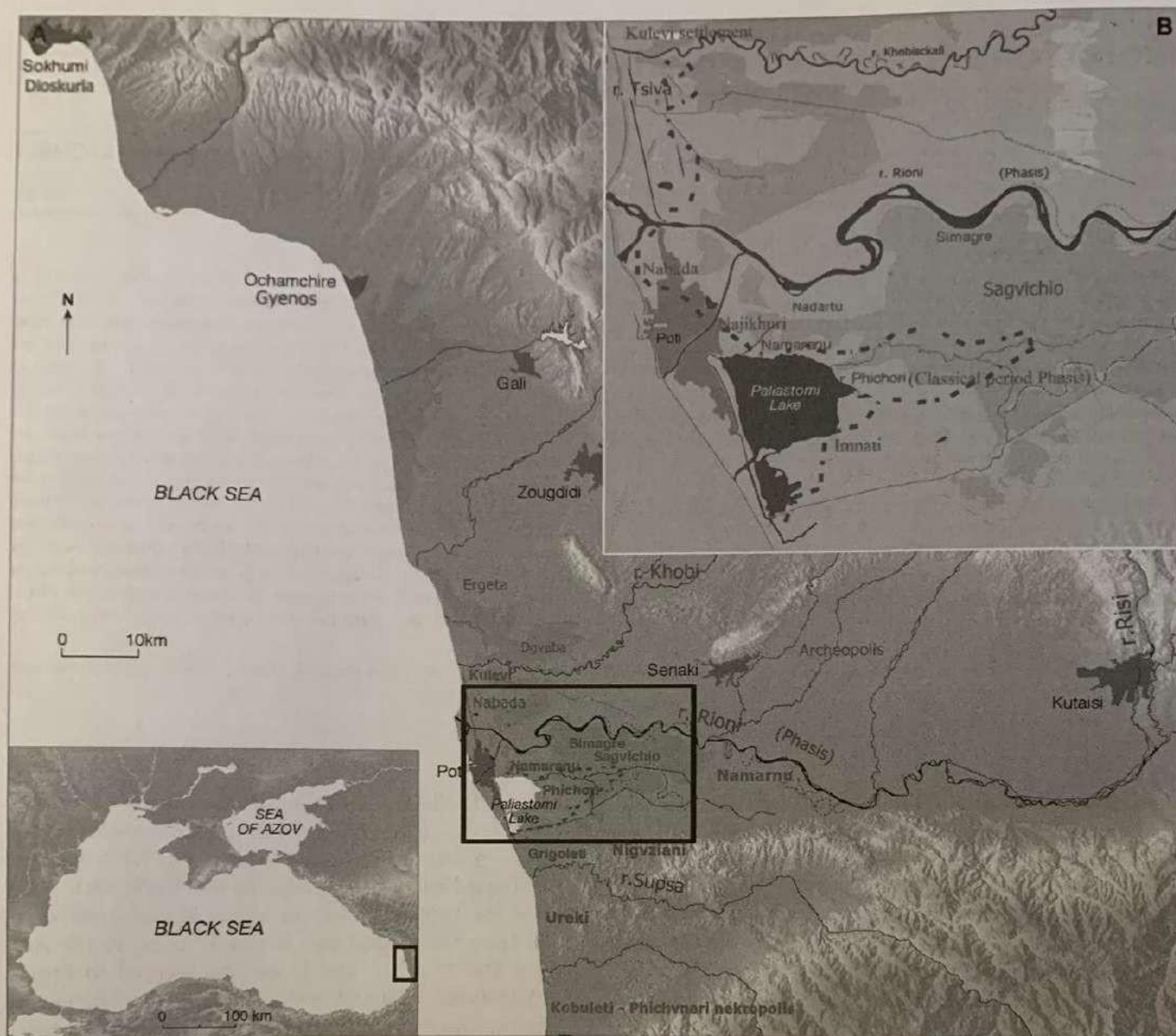


Fig. 1 – A: Map of Colchis and the distribution of archaeological sites mentioned in the text; B: The study area and Rioni estuary

four groups of these statues (PAPUASHVILI, 2010). Statues of a horse-riding goddess (holding a baby in her arms) are known only from Samos and Colchis. The Samos statue has been repeatedly published, thus I will mention it only briefly (LICHELI, 2008). Its origin was related to the Caucasian world (LORDKIPANIDZE, 2002). The geography of the distribution of the statues reveals that all four groups are only found in Colchis, which provides confirmation of Colchis's intermediary role.

Affinities between bull statuettes from Boeotia (THOMATOS, 2006) and statuettes from the Colchis burial grounds are also significant. Despite the different material used in their production (terracotta and bronze), the stylistic peculiarities are undoubtedly the same. Thus, according to small plastic samples, here we have a conceptual similarity not only with the Goddess/horsewoman in Greece and Colchis but also an iconographic resemblance with worshipping the powerful cult of a bull (about the possible interrelation between Colchis and Aegean World before Ionian colonization see: LORDKIPANIDZE, 2001; LICHELI, 2008;

LOMITASHVILI, 2003). Thus, Colchis did not appear as an unfamiliar region for the Greeks before the Ionian colonization. Existing archaeological and written materials do not allow the reasons for earlier relations to be precisely determined (LORDKIPANIDZE, 2002; TSETSKHLADZE and TREISTER, 1995);

2 - Phasis – problem of identification

During the mid 1st mill. BC, the Transcaucasia territory, including the Black Sea coast, was shaped by different political and economic events. The territory between the Black Sea and the Caspian Sea witnessed two large synchronous events, which finally divided this region politically and culturally into two parts: the Colchis coast and the rest of the territory located to the East. The first part was influenced by Greek colonization, which had appropriate consequences, and another appeared to be influenced by integrated Achaemenid empire and an increase of Achaemenid tendency was the logical result here as well (LICHELI, 2013).

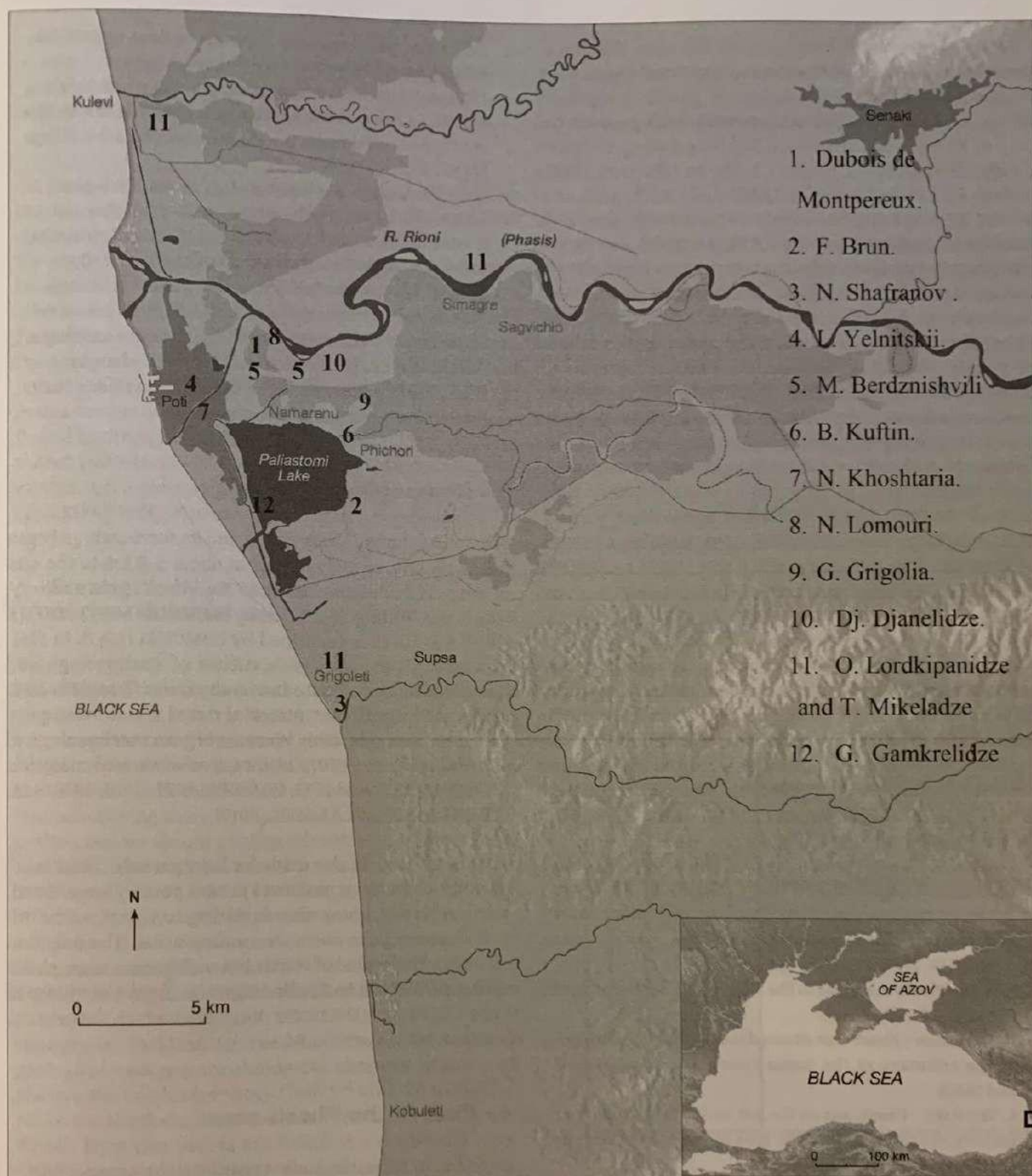


Fig. 2 – Twelve different opinions regarding the location of the city of Phasis.

These various events are reflected in archeological materials and written sources and have led to different interpretations (significantly contrasting opinions were discussed at the Tskaltubo and Vani symposiums: *Problems of the Greek Colonization of the Northern and Eastern Coasts of the Black Sea Region*, 1979; *The Demographic Situation in the Black Sea Region in the Period of the Great Greek Colonization*, 1981; *Local Ethno-Political Groups in the Black Sea Region in the 7th-4th Centuries BC*, 1988; *Le Pont-Euxin vu par les Grecs*, 1990; TSETSKHLADZE, 1992; GAGOSHIDZE, 1983; GVELESIANI, 2007; KNAUSS, 2006; LICHELI, 2007 and 2011; LORDKIPANIDZE, 1985). In any case, from the

beginning of Greek colonization and even earlier periods this coastal area was oriented towards Greece, as attest the archaeological findings. In comparison with other Black Sea coasts, the coastline of Colchis appeared less colonized. Based on written sources only three colonies were established there: Dioscourias (present Sokhumi and hinterland), Gyenos (present territory of Ochamchire) and Phasis. The location of Phasis is the most complicated, but it is generally accepted that it lies within a triangle, to the West bounded by the rivers Khobi and Supsa and to the East by the settlement of Simagre.



Many surveys have been carried out with the aim to understand Greek colonization and the cities' chronology, location and functional issues. Due to the complexity of the problem, special attention is here paid to the city of Phasis. For historical details relating to Phasis I suggest interested readers refer to the remarkable survey by LORDKIPANIDZE (2000 and 2010). Also, it is worth highlighting the small encyclopedic reference book by KACHARAVA and KVIRKVELIA (1991). Among the fundamental works of recent years, we can highlight the work of TSETSKHLADZE (1992) and GAMKRELIDZE (2009). However, despite the topic's popularity (GRIGOLIA, LICHEL and SHAVLAKADZE, 2013), many issues related to Phasis, including its location, remain unresolved. The location of Phasis is the major issue. This scientific problem was raised by DUBOIS DE MONTPEREUX (1839) and it is still significant today. It has been described in many works (the chronological list of the main publications are: BRUN, 1880; SHAFRANOV, 1885; YELNITSKIY, 1938; BERDZENISHVILI, 1942; MELIKISHVILI, 1966; LOMOURI, 1957; GRIGOLIA, 1973b; JANELIDZE, 1973; LORDKIPANIDZE and MIKELADZE, 1973; GAMKRELIDZE, 1987 and 2009; GRIGOLIA, LICHEL and SHAVLAKADZE, 2013).

Such a high number of publications reveals the great interest of scholars of different generations in trying to solve this problem. Of all the cases outlined, the region around the modern city Poti is one of the most likely location for Phasis. In total, twelve different opinions have been discussed regarding the city's location (GAMKRELIDZE, 2012). He lists the following hypotheses for the locations of Phasis (Fig. 2):

- Dubois de Montpereux - Roman Phasis lay between Chaladidi village and Poti town, namely, south of the latter (present-day airport). Here, he had found ruins of a fortress and thought that it was Arrian's Phasis (DUBOIS DE MONTPEREUX, 1839);
- F. Brun - Phasis town was in the south - east section of lake Paliastomi (BRUN, 1880);
- N. Shafranov - Phasis was situated south of lake Paliastomi, at the tributary of the Supsa river (SHAFRANOV, 1882 and 1885);
- L. Yelnitskii - Phasis was on the left bank of the Rioni river estuary, approximately in the area of the present day sea port (YELNITSKII, 1938);
- M. Berdznishvili - Phasis lay on the left bank of the Rioni River, near *Patara* (small) Poti. Regarding the Phasis of Arrian's times, he agrees with Dubois de Montpereux (BERDZNISHVILI, 1942);
- B. Kuftin - Arrian's Phasis lays on the estuary of the Pichori river, on the banks of Paliastomi lake (KUFTIN, 1950);
- N. Khoshtaria - the Phasis of Greek and Roman-Byzantine periods was situated in the area of present-day Poti town (GAMKRELIDZE, 2012);
- N. Lomouri agrees with Dubois de Montpereux but believes that it is not the Phasis which is mentioned by Arrian, but rather the Phasis of Agathias' times (6th c. AD) (LOMOURI, 1957).
- G. Grigolia - Phasis should be searched for on the east side of Paliastomi lake, where the Pichori river flows into the lake (GRIGOLIA, 1973b);
- Ch. Janelidze - Phasis town should be searched for along the Rioni river, at a 6 km distance from the sea, in the territory around *Patara* (small), Poti and Chaladidi village (JANELIDZE, 1973);
- O. Lordkipanidze and T. Mikeladze - Phasis corresponds to archaeological sites on the estuary of the Rioni river, around *Kvemo* (lower), Chaladidi and the coastal villages of Grigoleti and Kulevi (LORDKIPANIDZE and MIKELADZE, 1973);
- G. Gamkrelidze - the 3rd - 7th century Phasis is the same as "Natekhebi" settlement found in the southern part of Poti town, on the western side of Paliastomi lake. According to GAMKRELIDZE (1987), palaeoenvironmental changes along the Kulevi, the Rioni, Pichori and Supsa rivers led to Phasis being relocated several times.

3 - Archaeological findings

The archaeological study of Poti and its surroundings began relatively late, namely in 1952 at about 5-6 km to the east of the port of Poti, in the area "Najikhuri". This territory, which was initially identified by DUBOIS de MONTPEREUX (1839) was further examined by LOMOURI (1957). In 1961, archaeologists from the Institute of Archaeology and Ethnology of Tbilisi State University (head T. MIKELADZE) discovered significant material dated to late Antiquity and later periods. The West Georgian Archaeological Expeditions in 1969-1970 also excavated various materials (GRIGOLIA, 1973 and 2010; GAMKRELIDZE, 1987; GRIGOLIA, LICHEL and SHAVLAKADZE, 2013).

Discoveries were also made for later periods: "Najikhuri" Byzantium and late medieval period pottery were found, along with amphorae sherds dating to the 1st - 8th c. AD in Paliastomi Lake and surrounding areas. The only find related to the period of Phasis is a well-known silver phiale with a dedication to Apollo Hegemon (from the village of Zubovka; outside the limits of Colchis), which dates to the last quarter of the 5th c. BC.

4 - Peat in the Phasis area

In order to solve the issue regarding the accessibility of the lake and sea gulf and to understand the evolution of the coastal landscapes, a survey of existing peat layers is important (see NICHOLAS *et al.* 2011). Peat is formed in humid areas. In coastal areas, peat is a reliable sea-level indicator. Imnati peat deposit is located on the south bank of Lake Paliastomi, where the maximum thickness of the peat layer is 12 m. The deepest layer of the peat has yielded a radiocarbon determination of 5825 ± 125 years, while the youngest peat layer has been dated to 2100 ± 150 years (JANELIDZE, 1973). JANELIDZE described the stratigraphy of the Imnati peat deposit as follows:

- 0 - 4.75 m sphagnum peat with marsh-plant remains (sedge, *calamagrostis* and others). The rate of peat decay is 5-15 %;
- 4.75 - 8.00 m grassy sphagnum peat. The rate of peat decay



is 23-35 %;

- 8.00 – 15.50 m sphagnum sedge peat (sedge, reed of *calamagrostis*). The rate of decay is 35-40 %;
- 10.50 – 11.30 m grassy sphagnum peat;
- 11.30 – 40.00 m lagoon clay.

These data indicate that the plain south of the Paliastomi Lake has not been flooded for the last 6000 years (JANELIDZE, 1973). The situation is similar to the north with the mouth of the river Phasis (Nabada peat residue). The distance from the Nabada deposit's southern edge to the north bank of the Paliastomi lake (which was freshwater until 1924, when the small channel between the Lake and the sea was constructed) does not exceed a maximum length of 6-7 km. Peat deposits are not observed in this narrow isthmus demonstrating that the river Phasis flowed in this area for 2000 - 4000 years and merged with the sea (JANELIDZE, 1973). Ptolemy (2nd c. AD) describes the existence of a marine bay here (*Geographic manual* V, VI, 1; LORDKIPANIDZE, 2010), as well as Arrian (see below).

5 - Stratigraphy of the settlement of the "colonial area"

It is obvious that any serious colonization requires previous knowledge of both the land to be settled and of the people who inhabited it (GRAHAM, 2001). In this context, it has been suggested that Phasis was already settled before the arrival of the Greeks (GAMKRELIDZE, 2012). Potentially, this is possible because there is information regarding Heniochoi living there (Pomp. Mela, I, 108) (DE BOER, 2007). In this case, we should distinguish between "colonization" and "establishment" because this process continued from 550 BC until the 4th c. BC (FIGUEIRA, 2008). The decision of Milesian Greeks to establish a colony near the biggest river of Colchis, Phasis, was logical. However, geomorphology changes, meaning that Phasis' location probably modified during the Classical, Hellenistic, Roman and Byzantine periods (KACHARAVA and KVIRKVELIA, 1991). These changes of location (if they occurred) were due to geological and geomorphological changes which took place in the Colchis territory from 2nd mill. BC until 5th c. AD. In the Northern Black Sea, architectural remains and burials from this period are found at a depth of 2 - 3 m. These events were followed by changes in the hydrological network of the coastal area, central Colchis and its adjacent territory (JANELIDZE, 1973; LORDKIPANIDZE, 2000), which is confirmed by the fact that one another colony established in Colchis, Dioscurias, is presently below the sea level, in addition to the early layer from the settlement of Kulevi. Thanks to recent archeological excavations of the Kulevi settlement, which was located on the Khobi river bank (river Kharient of Arrian), near the confluence, the chronology of geomorphological events has been elucidated. Kulevi settlement was on a 1.6 m high artificial accumulation, on which four cultural layers with a total thickness of 3 m have been described (LICHELI, PAPUASHVILI, and PAPUASHVILI, 2000; PAPUASHVILI, PAPUASHVILI, 2014). The stratigraphy is as follows:

- 4th layer – second half of 2nd mill. BC (thickness 0.5 m) was located 1.4 m below current sea level;
- 3rd layer – end of 2nd mill. BC and beginning of 1st mill. BC, was located below sea level;
- 2nd layer – first quarter of 1st mill. BC.;
- 1st layer – 19th century.

Located above sea level, the remains of log plaster structure, bronze, iron, clay productions were revealed. Furthermore, the remains of metallurgy production (moulds, ingots, pipes) and items with a ritual purpose were found (altars, double - ram and dolphin representations; remains of wheat, millet, nut, acorn and chestnut).

A clearer picture of the geomorphological changes during the period of colonization was revealed after excavations of the settlement on the right bank of the river Tsiva. Its lower horizon corresponds to the 6th - 5th c. BC surface and is located 0.8 m below sea level. Fragments of Ionian black glazed vessel and Chian amphorae were revealed in this stratum (PAPUASHVILI and PAPUASHVILI, 2014). Fragments of 4th - 3rd c. BC Sinopian and Colchian amphorae were discovered in the upper stratum (Fig. 3). Geological surveys carried out in 2000-2007 (300 boreholes down to a depth of 50 m) showed that layers of 6th - 4th c. BC are at a depth of 3 m.

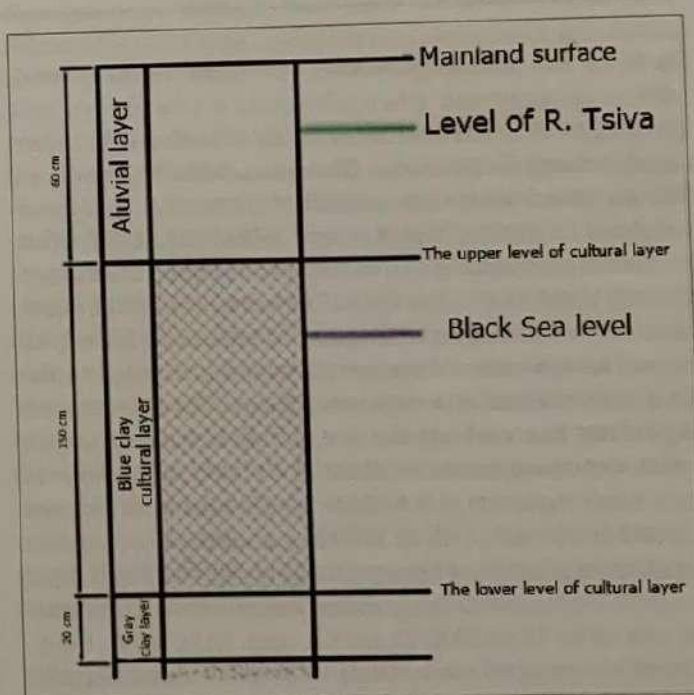


Fig. 3 – Stratigraphy of Tsiva settlement – After Papuashvili and Papuashvili, 2014.

Thus it is obvious that the 1st mill. BC settlement of Kulevi was constructed on an artificial elevation (Fig. 4), which is common for Colchis. As a result of survey works carried out by my expedition in Grigoleti, black glazed pottery from the 5th-4th c. BC was discovered in the layer above sea level, which proves that the settlement of this period was not flooded by the sea. The same can be said of Kobuleti-Phichvnari necropolis from the 5th - 4th c. BC (Greek and Colchian), which also is located above the Black Sea level (VICKERS and KAKHIDZE, 2004;

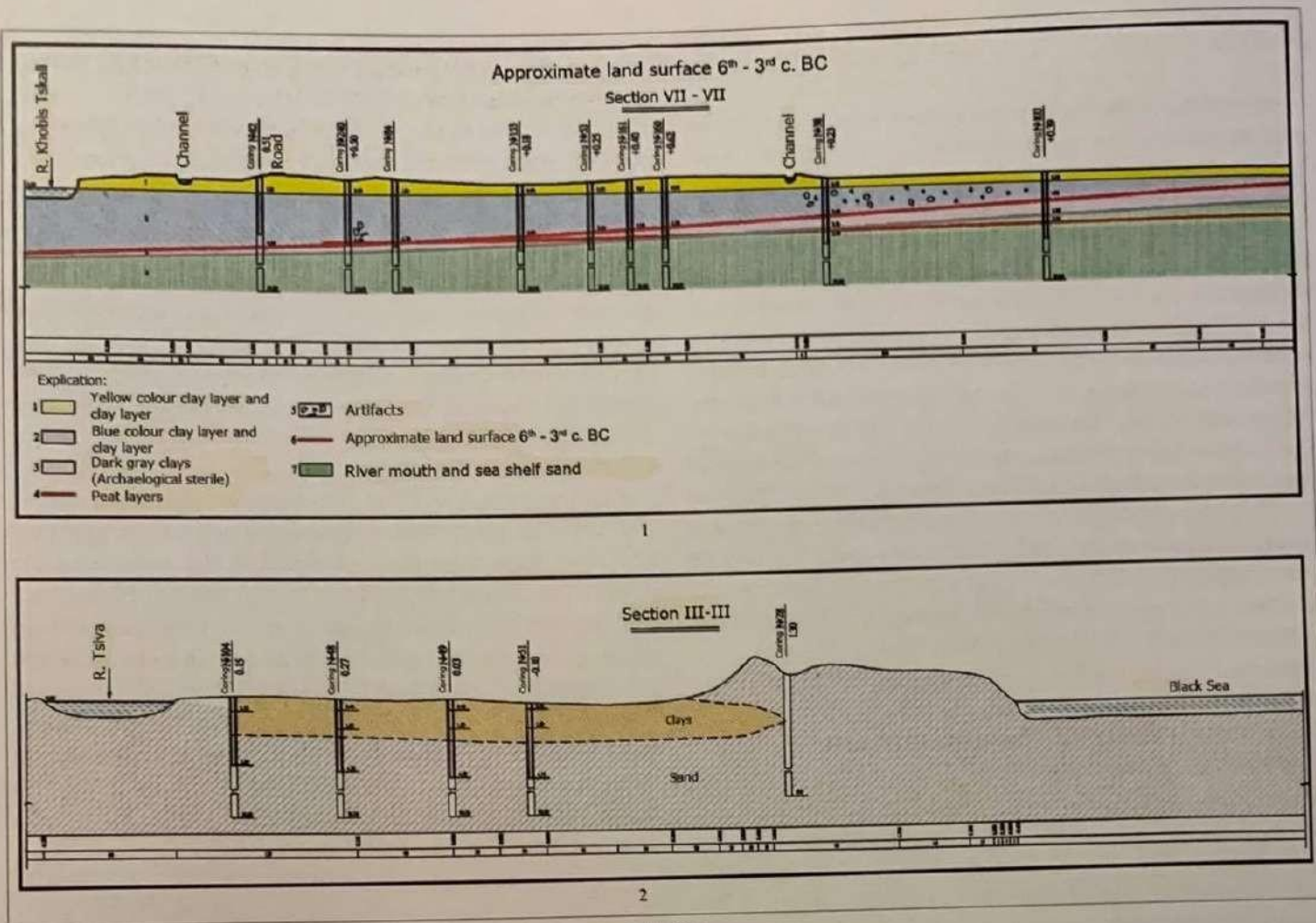


Fig. 4 – Stratigraphy of Kulevi settlement – After Papuashvili and Papuashvili, 2014.

KAKHIDZE and TAVAMAISHVILI, 2007). Also is another Greek colony in Colchis – Dioscourias (Sebastopolis in Roman times) which, as a result of tectonic activity, has subsided (KHROMOVSKIKH and NIKONOV, 1984). Two historical earthquakes need to be considered (50 AD and 400 AD, VARAZASHVILI, TSERETELI and TSERETELI, 2011), which twice destroyed Dioscourias/Sebastopolis. In both cases, the epicenter of the earthquake was located to the South, in the sea at a depth of 10 km. The information regarding this earthquake are as follows: date = 50 AD; epicenter coordinates: $\varphi = 42.90^\circ$, $\lambda = 41.00^\circ$; depth – $h = 10$ km; magnitude $-M_s = 5.5$ ($DM = \pm 0.5$) intensity at the epicenter $-I_0 = 8$ ($DI_0 = \pm 1$). It is clear that the drowning of a significant portion of Sebastopolis was the result of rapid seismic dislocation. Such catastrophes were numerous in the area (KHROMOVSAKIKH and NIKONOV, 1984). From this point of view, it has to be mentioned that the Northern and Southern parts of the presumable area of Phasis were located within the zone of maximum damage caused by large historical earthquakes (Fig. 5). The altitude of submerged cultural layers from the 1st mill. BC has been interpreted as being linked to tectonic activity (FOUACHE *et al.*, 2012). In 1000 BC, sea level was located around -2 m below its present position in S5 (eastern part of Anapa Spit S5), around -4 m in S3 (Choushka Spit (S3) and around -9.5 m in S2 (Burgaz Spit (S2) and S4 (Western Anapa Spit (S4), (FOUACHE *et al.* 2012).

These data confirm that the subsidence rates vary on the peninsula (FOUACHE *et al.* 2012). Furthermore, in relation

to the Taman Peninsula, it is clear that each area shows a unique relative sea-level history. A synoptic view shows that the differences can best be interpreted by tectonics and compaction (FOUACHE *et al.* 2012). Considering these facts, possible transgression (transgression – regression patterns are still under discussion: BRÜCKNER *et al.* 2010) and also historical earthquakes, we can assume that analogous events took place in Colchis which explains the underwater position of many Greek cultural layers. Although it must be underscored that settlement levels (= elevation of living levels) in the 1st mill. BC varied from site to site.

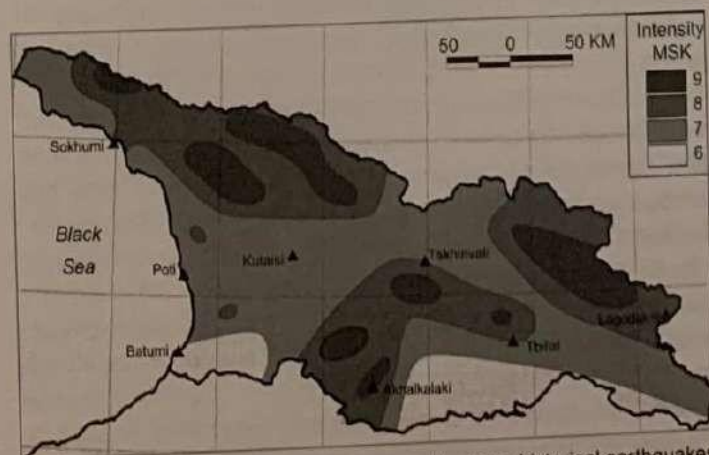


Fig. 5 – Summary map of the damage caused by large historical earthquakes in Georgia – After Varazashvili, Tsereteli, Tsereteli, 2011.

6 - The coastline, lake and city of Phasis

Without doubt, the topography of the city of Phasis was analogous to other Colchian settlements. The settlement of Sagvichio is a wonderful example of this (distance from coastline 20 km to the East), similar to the Simagre settlement (distance from coastline 15 km to the East). Both fell within the economic sphere of Phasis, as attested by the great number of "urban-type" Greek imports. Three cultural layers were discovered in the Simagre settlement, which include a quite narrow chronological frame: the end of the first half of 6th c. BC and the beginning of the second half of the 5th c. BC. The level of the floor of the third layer is found 1.5-2 m below the current r. Rioni (Phasis) terrace (MIKELADZE, 1978). Regarding the Sagvichio settlement, it is distinguishable thanks to its large size and its location in the r. Rioni-Pichori waterlogged nature reserve. The Sagvichio settlement consists of five hills and is surrounded by elevated agricultural land; the total area consists of a thousand hectares (unfortunately satellite coordinates and the real size have not yet been published (GRIGOLIA, 2010). The central hill of the monument is oval and 7-8 m high (diameter 60 m). Only a few survey trenches were undertaken on this hill and the following artifacts were discovered along with local production items: fragments of kylix (first half of the 6th c. BC); skyphos fragments; Lesbian, Chian, Thasian amphorae of 5th c. BC. These facts indicate that it is incorrect to consider all the settlements of Colchis as one united group. The location of the city of Phasis does not confirm the view of its total submergence. Furthermore, adjacent to Lake Paliastomi (to the North East) between the rivers Rioni and Pichori, on the left bank of the Nadartu channel (just 5 km distance from Simagre settlement to the South West), an archaeological site contemporary to Simagre and with similar characteristics ("urban type" Greek import) was discovered. 700 m West of this site, there is an unexcavated hill, Namaranu, with a height of 1 m a.s.l., where typical plaster of Colchian buildings can be seen. The distance of these sites from the modern coastline is 8 km. One important fact should be considered when discussing Phasis' geomorphological environment – when Strabo says "On Phasis is situated a city bearing the same name, an emporium of the Colchi, which is protected on one side by the river, on another by a lake, and on another by the sea" (XI, 2, 17) by this we outline the geomorphological situation of the 1st c. BC- 1st c. AD (as well as in 6th c. See: Agathias "there is a lake there". *Georgica*, III, 20). Based on stratigraphic and geomorphological data, it is presumed that 2000-3000 years ago no other lake other than Paliastomi existed in the Poti adjacent territory (JANELIDZE, 1973). But Pseudo-Scylax, (4th c. BC; KAUKHCHISHVILI, 1967) does not mention Paliastomi lake (Ps.-Sc., 81), which suggests that Phasis was not situated near a lake and that the lake does not have a significant meaning for the early location of Phasis. The background for such a suggestion is that, in comparison to other settlements of Colchis, Pseudo-Scylax pays attention to each detail, for example, he indicates the "barbarian city of Medea", which is

attributed to the Namarnu settlement (GRIGOLIA, 1973a) or Archeopolis (KAUKHCHISHVILI, 1967). Furthermore, along with a "Barbarian city" which is located within the country, he also mentions the river Ris. It is obvious that he pays attention to water resources, even small-sized rivers existing in the Colchis territory, but – as has been mentioned – he does not indicate a lake. The absence of a lake near the city of Phasis became clearer in the 2nd c. AD. Arrian, who seems to know a great deal about Phasis, does not mention a lake at all. This is odd as he describes reinforcement works and hydrological networks in detail. The most satisfactory explanation is that during Arrian's time a lake did not exist near Phasis, although there was a port. (Arr. 9: "I thought proper to carry from the double ditch, that surrounds the wall, another ditch, as far as the river, which may include both the harbor, and the buildings, that lie beyond the walls of the fortifications"). Furthermore, Ptolemy indicates the existence of a seaway here ("Colchis is bounded by part of Pontos-Euxinos from the West from the river Koraki up to the Phasis cavity", *Geographic manual* V, VI, 1; LORDKIPANIDZE, 2010). The geomorphological environment at the time of Phasis' foundation, and in later periods too, was similar to Ptolemy's times.

In the context of discussion about relative sea-level changes since 7500 BP, FILIPOVA-MARINOVA (2007) has reconstructed three hypothetical sea-level highstands above present sea level, including one at 4 m a.s.l. at 3730 cal BP, which chronologically corresponds to the period of Greek colonization. In this case the results of the investigation of coastal changes are as follows: change of sea level (5th c. BC- 2nd AD?) by +1 m provokes a coastal shift of 10-13 km to the East; + 2 m of 15 km; + 4 m of ca. 25-27 km. It is significant that estuary formation seems to be more frequent in the Pichori basin. Accordingly, during the period when the estuary existed (at least from the classical era until Arrian) it was within the Pichori basin. The existence of a lagoon coincides with the boundaries of the peat deposit. A decrease in sea level around 310 cal BP (according to FILIPOVA-MARINOVA's hypothesis, the peak of the sea level at this time was 1 m a.p.s.l) for the territory of Colchis is possible (LORDKIPANIDZE, 2000). It is indeed extremely challenging to use Bulgarian data to reconstruct the geomorphology of Colchis.

Sagvichio settlement existed between the rivers Rioni and Pichori, approximately 17 km east of the present coastline. According to the characteristics of Sagvichio settlement, the location of the early city of Phasis seems to be clear – this is Sagvichio settlement or territory east from Sagvichio.

It is believed that the city of Phasis was situated on the left bank of the river Phasis (KACHARAVA and KVIRKVELIA, 1991; LORDKIPANIDZE, 2010; GAMKRELIDZE, 2009). Analysis of written sources shows that different interpretations are possible. It should be mentioned that when the term "Phasis" is used in the masculine context (BRAUND, 1994), it always stands for a

river. Therefore, there is an indication about the location of the city of Phasis in the written source of Arrian:

The Statue of the Goddess Phasiane is placed to the left of the entrance into the Phasis (The Periplus of the Euxine Sea, 7).

The same can be said with regards to Anonymus (5th c.):

Hellenic town so called Phasis is located on the left side of the river Phasis which was established by Miletus.

Other authors (except Strabo, mentioned above) do not provide any significant information regarding the river and city of Phasis, for example Pindar (c. 522 – c. 443 BC), in the *Pythian Odes* (4.205):

Than the battle-lines of the loud-thundering winds. But that voyage of the demigods put an end to them. And then the Argonauts came to Phasis, where they clashed with the dark-faced Colchians in the realm of Aeetes himself. And the queen of sharpest arrows brought the dappled wryneck from Olympus, bound to the four spokes.

Or Ammianus Marcellinus (4th c. AD) (XXII. 8.24):

Beyond these territories are the populous districts of the Camaritae, and the Phasis in impetuous course borders on the Colchians, an ancient race of Egyptian origin. There, among other cities, is Phasis, which gets its name from the river, and Dioscurias, well known even to this day, said to have been founded by Amphitus and Cercius of Sparta, the charioteers of Castor and Pollux, and founders of the nation of the Heniochi.

Or Hippocrates ("About Airs, Waters and Places", XV):

As to the inhabitants of Phasis, their country is fenny, warm, humid, and wooded (...) for their dwellings are constructed of wood and reeds, and are erected amidst the waters; they seldom practice walking either to the city or the emporion, but sail about, up and down, in canoes constructed out of single trees, for there are many canals there. They drink the hot and stagnant waters, both when rendered putrid by the sun, and when swollen with rains. The Phasis itself is the most stagnant of all rivers, and runs the smoothest.

Even if the *emporion* could be identified with Phasis, there is no indication of location and its position relative to the river/lake.

Based on the above mentioned evidence, I suggest that the early city of Phasis was situated on the northern bank of the river Phasis. Nonetheless, at a particular time (6th c. AD), for Agathias the Scholasticus (*Georgica*, III, 20) the city appeared to be south to the river due to the migration of the river. This is the result of the natural mobility of the channel of Rioni (LORDKIPANIDZE, 2000).

Conclusion

Based on the above opinions, it is possible to state the following:

- The assumption that the river Phasis runs into today's Pichori riverbed is acceptable. In addition to the geomorphological traits, this opinion is supported by the 6th-5th c. BC concentration of archaeological sites on both banks of the Pichori (personal communication with G. Grigolia) and also, between the Pichori and Rioni rivers.
- The early city of Phasis was situated on the Northern bank of the river Phasis (Pichori), though at a particular time the city appeared to be South of the river.
- The city of Phasis was not established near a lake and Lake Paliastomi is not significant for the location of the early city. Presumably, Lake Paliastomi did not exist at that time in the same place as it is today, as mentioned above. The lake is not described in any written sources (classical and Hellenistic era). Despite the fact that, according to Strabo, the city of Phasis was related to the Lake Paliastomi area (XI, 2), it is not impossible that a lake existed north of the river Phasis and indeed Strabo evoked this fact.
- Despite of the theory about the marine transgression, the early city of Phasis (6th - 4th c. BC) can be attributed to Sagvichio settlement.

References

- BRAUND D., (1994), *Georgia in Antiquity. A History of Colchis and Transcaucasian Iberia, 550 BC - 562 AD*, Clarendon Press, Oxford, 359 p.
- BERDZENISHVILI M. D., (1942), To localization of city Phasis, *IOTSU*, XXIII, p. 19-20 (in Georgian).
- BRÜCKNER H., KELTERBAUM D., MARUNCHAK O., POROTOV A., VÖTT A., (2010), The Holocene sea level story since 7500 BP – Lessons from the Eastern Mediterranean, the Black and the Azov Seas, *Quaternary International*, 225, p. 160-179.
- BRUN F. K., (1880), *Black Sea littoral*, II, Odessa, 250 p. (in Russian).
- DE BOER J. G., (2007), The earliest possible date of Greek colonization along the western Pontic coast, in LAZARENKO I. (ed.), *Ancient Civilizations and the Sea, International Conference In Honorem Annorum LXX Mihaili Lazarov Acta Musei Varnaensis*, 5, Varna, p. 123-142.
- DILLON M., GARLAND L., (2010), *Ancient Greece: Social and Historical Documents from Archaic Times to the Death of Alexander the Great*, 2, 624 p.
- DUBOIS DE MONTPEREUX F., (1839), *Voyage autour du Caucase*, 3, Gide, Paris, p. 62-81.

- FIGUEIRA T., (2008), Colonization in the Classical period, *Greek colonization: an account of Greek colonies and other settlement oversea*, 2, Brill, Leiden, p. 427-523.
- FILIPOVA-MARINOVA M., (2007), Archaeological and Paleontological evidence of climatic dynamics-sea – level change and costline migration in the Bulgarian sector of the circum-pontic region, in YANKO-HOMBACH V., GILBERT A. S., PANIN N., DOLUKHANOV P. M. (eds), *The Black Sea Flood question: Changes in Coastline, Climate, and Settlement*, Amsterdam, Springer, p. 453-481.
- FOUACHE É., KELTERBAUM D., BRUCKNER H., LERICOLAIS G., POROTOV A., DIKAREV V., (2012), The Late Holocene evolution of the Black Sea – a critical view on the so called Phanagorian regression, *Quaternary International*, 266, p. 162-174.
- GAGOSHIDZE I., (1983), *Kartli (Iberia) in the 5th-1st centuries B.C.*, PhD Thesis, Tbilisi, p. 24.
- GAMKRELIDZE G., (1987), Researches of underwater-archaeological expedition on the area of Poti, *Matsne*, 1, p. 97-117 (in Georgian).
- GAMKRELIDZE G., (2009), The Land of Colchis and city of Phasis, *Iberia-Colchis*, 5, p. 75-194 (in Georgian).
- GAMKRELIDZE G., (2012), *Researches in Iberia-Colcholog*, Broundm D., Tbilisi, 345 p. (in Georgian).
- GRAHAM A. J., (2001), *Collected Papers on Greek Colonization*, Leiden, Brill, 416 p.
- GRIGOLIA G. K., (1973a), Location of great Barbarian city, *Dzeglis Megobari*, 33, p. 50-58 (in Georgian).
- GRIGOLIA G. K., (1973b), The localization of Phasis, *Proceedings of the conference dedicated to the problems of city Phasis localization*, p. 36-55 (in Georgian).
- GRIGOLIA G. K., (2010), The settlement of Sagvichi, *Cultural and Historical researches*, Tbilisi, p. 75-79 (in Georgian).
- GRIGOLIA G. K., LICHELI V., SHAVLAKADZE C., (2013), Poti archaeological expedition, *Interdisciplinary archaeology*, II, Tbilisi, p. 8-27.
- GVELESIANI M., (2007), Supreme Deity of Armazi-Kartli, *Iberia-Colchis*, 3, p. 33-50.
- JANELIDZE C., (1973), Geographical aspect of Phasis localization, *POSCDTLOP*, p. 16-19 (in Georgian).
- KACHARAVA D., KVIRKVELIA G., (1991), *Black Sea towns and cities in antiquity*, Tbilisi, p. 289-290, (in Georgian).
- KAKHIDZE A., TAVAMAISHVILI M., (2007), Works held on Hellenistic Cemetery of Pichvnari in 2003-2004 years, *Works of the Archaeological Museum of Batumi*, 4, p. 43-69 (in Georgian).
- KAUKHCHISHVILI T., (1967), *Reports of Greek writers about Georgia*, 1, Tbilisi, p. 36 (in Georgian).
- KHROMOVSKIKH V., NIKONOV A., (1984), *Following strong earthquakes*, Nauka, Moscow, 144 p.
- KNAUSS F., (2006), Ancient Persia and the Caucasus, *Iranica Antiqua*, XI, p. 79-118.
- KUFTIN B. A., (1950), *Materials to archaeology of Kolkhis*, I, Tbilisi, p. 116 (in Russian).
- liche V., (2007), New discoveries in Colchis, an interpretative version. The Argonautica and World Culture, *Phasis, Greek and Roman Studies*, 10, p. 109-115.
- liche V., (2008), The statue of Colchian knight on the Samos island – accidental or reasonable, in VICKERS M., BRAUND D., PAPUCI-WLADYKA E., BODZEK J. (eds), *Pontika, Recent research on the Northern and Eastern Black sea in Ancient Times*, BAR, Oxford, p. 49-50.
- liche V., (2013), Achaemenids and Caucasian Iberia, in KLUCHNIKOV W. (eds), *Pricheronomir'ie v Antichnuui I srednevekovie epoxi*, Rostov-na-Donu, p. 169-174.
- liche V., PAPUASHVILI R., PAPUASHVILI N., (2000), Preliminary results of excavations at the Kulevi location, *International scientific conference: Archaeology and Ethnography of Caucasus, summary reports*, Baku, 88 p. (in Russian).
- LOMITASHVILI D., (2003), *Central Colchis in 8th c. BC and 4th c. AD*, *Tsikhe-Goji, Archeopolis Nokalakevi*, PhD Thesis, Tbilisi, p. 7-20 (in Georgian).
- LOMOURI N. I., (1957), From Historical Geography of Ancient Colchis, *HOAH*, 4, p. 96-110 (in Georgian).
- LORDKIPANIDZE O., (1985), Some issues from the history of Kartli in the 5th-1st centuries B.C., *Vestnik Drevnei Istorii*, Tbilisi, p. 98.
- LORDKIPANIDZE O., (2000), *Phasis. The river and city in Colchis*, Stuttgart, 145 p.
- LORDKIPANIDZE O., (2001), The Golden Fleece: myth, euhemeristic explanation and archaeology, *Oxford Journal of Archaeology*, 20, p. 1-38.
- LORDKIPANIDZE O., (2002), *Down of ancient Georgian civilization*, Iv. Javakhishvili Tbilisi State University, Tbilisi, 207 p. (in Georgian).
- LORDKIPANIDZE O., (2010), *Ancient Georgia (Kolkhis and Iberia) in Geography of Strabo*, Tbilisi, p. 187-194 (in Georgian).
- LORDKIPANIDZE O., MIKELADZE T. K., (1973), Problem of historical and archaeological researches of Phasis, *POSCDTLOP*, p. 27-33, (in Russian).
- MELIKISHVILI G. A., (1966), Naming cities of Phasis and issue of ethnic staff of population of ancient Kolkhis, *Vestnik Drevnei Istorii*, 1, p. 82-86 (in Russian).
- MIKELADZE T., (1978), Archaeological investigation in the Rioni valley, *Materials towards the history of ancient Phasis*, Tbilisi, p. 21-92.
- MIKELADZE T., (1995), Grosse Kollektive Grabgruben der frühen Eisenzeit in Kolchis, *Archäologischer Anzeiger*, 1, p. 2-22.
- NICHOLAS W. A., CHIVAS A. R., MURRAY-WALLACE C. V., FINK D., (2011), Prompt transgression and gradual salinisation of the Black

- Sea during the early Holocene constrained by amino acid racemization and radiocarbon dating, *Quaternary Science Reviews*, 30, p. 3769-3790.
- PAPUASHVILI R., (1989), *For comparable chronology of burial pits, according to weapons*, PhD Thesis, Tbilisi, p. 43-57 (in Georgian).
- PAPUASHVILI R., (1998), *Comparable chronology of Kolkhis collective burials according to weapons*, Tbilisi, 39 p. (in Georgian).
- PAPUASHVILI R., (2004), Collective burial pit 4 of Ergeta II burial ground, *Journal of Georgian Archaeology*, 1, p. 152-154 (in Georgian).
- PAPUASHVILI R., (2010), Anthropomorphic images from Tsaishi, *Cultural and Historical researches*, Tbilisi, p. 49-51 (in Georgian).
- PAPUASHVILI R., PAPUASHVILI I., (2014), *Paleoregime of Black sea and its influence on settlements of Kolkhis (III-I millennium B.C.)*, Mvignobari, Tbilisi, p. 73-74 (in Georgian).
- SHAFRANOV N., (1882), Formation of Paliastomi and source of Kapar-chai, *Collection of materials for description locals and tribes of Caucasus*, Tbilisi, p. 43-58 (in Russian).
- SHAFRANOV N., (1885), Formation of Rioni during historic period, in *Collection of materials about Kutaisi*, Kutaisi, first Edition, p. 45-58, (in Russian).
- THOMATOS M., (2006), *The final revival of the Aegean Bronze Age: a case study of the Argolid, Corinthia, Attica, Euboea, the Cyclades and the Dodecanese during LH III C. Middle*, BAR, Oxford, 346 p.
- TSETSKHLADZE G., (1992), Greek Colonization of the Eastern Black Sea Littoral (Colchis), *Dialogues d'Histoire Ancienne*, 18, 2, p. 223-258.
- TSETSKHLADZE G., TREISTER M., (1995), The Metallurgy and Production of Precious Metal in Colchis before and after the arrival of the Ionians (Towards the problem of the reasons for Greek colonization), *Bulletin of Metals Museum*, 24, p. 1-32.
- VARAZASHVILI O., TSERETELI N., TSERETELI E., (2011), *Historical Earthquakes in Georgia (up to 1 900)*, *Source Analyses and Catalogue Compilation*, Tbilisi, 22 p.
- VICKERS M., KAKHIDZE A., (2004), Pichvnari I: Results of Excavations Conducted by the Joint British-Georgian Pichvnari Expedition 1998-2002, in VICKERS M., KAKHIDZE A. (eds.), *Greeks and Colchians on the East Coast of the Black Sea*, Oxford, Batumi, p. 6-123.
- YELNITSKIY L.A., (1938), From Historical geography of ancient Kolkhis, *Herald of Ancient History*, 2, p. 307-320 (in Russian).

Literary sources

- Agathias, *Georgica*, Tbilisi, 1936.
- Ammianus Marcellinus, on Perseus Digital Library: [www.perseus.tufts.edu].
- Arrian of Nicomedia, *The Periplus of the Euxine Sea*, translated by William Falconer on [https://en.wikisource.org/wiki/The Periplus of the Euxine Sea].
- Hippocrates, *About Airs, Waters and Places*, translated by Francis Adams, online: [https://archive.org/details/b29005218].
- Pindar I, *Olympian Odes, Pythian Odes*, edited and translated by William H. Race, LCL 56. Harvard University, 2002.
- Ptolemy, *Geographic manual*, translated by N. LOMOURI, in *Materials for the History of Georgia and Caucasus*, 32, 1955, Tbilisi.
- Strabo, *Geographica*, Tbilisi, 1958.